RELATIVE FINITE PRESENTATION OF A MODULE IS INDEPENDENT OF THE PRESENTATION AND PASSES TO FINITELY PRESENTED QUOTIENTS OF THE ALGEBRA

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement span contains the author's three enumerated conditions and the closing consequence, delivered verbatim. The three references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Finite presentation of a module relative to a base ring, via a polynomial presentation. the conditions at statement lines 20928-20933. Is the property whose presentation-independence and quotient-invariance are established.
- Finiteness of a finitely presented module over a subring, and finite extensions of finite presentation. invoked at proof lines 20953-20954 and 20957. Are the two criteria that relate the conditions across the diagrams.

Source reference. Original lemma 0564: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6697–6697. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0561: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6694–6694. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $R \to A$ be a ring map of finite type. Let M be an A-module. The following are equivalent*

1. *for some presentation $\alpha : R[x_1, \ldots, x_n] \to A$ the module M is a finitely presented $R[x_1, \ldots, x_n]$-module,*
2. *for all presentations $\alpha : R[x_1, \ldots, x_n] \to A$ the module M is a finitely presented $R[x_1, \ldots, x_n]$-module, and*
3. *for any surjection $A' \to A$ where A' is a finitely presented R-algebra, the module M is finitely presented as A'-module.*

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In this case M is a finitely presented A-module.

Proof. If $\alpha : R[x_1, \ldots, x_n] \to A$ and $\beta : R[y_1, \ldots, y_m] \to A$ are presentations. Choose $f_j \in R[x_1, \ldots, x_n]$ with $\alpha(f_j) = \beta(y_j)$ and $g_i \in R[y_1, \ldots, y_m]$ with $\beta(g_i) = \alpha(x_i)$. Then we get a commutative diagram

$$\begin{array}{ccc} R[x_1, \ldots, x_n, y_1, \ldots, y_m] & \xrightarrow[y_j \mapsto f_j]{} & R[x_1, \ldots, x_n] \\ \downarrow{\scriptstyle x_i \mapsto g_i} & & \downarrow \\ R[y_1, \ldots, y_m] & \longrightarrow & A \end{array}$$

Hence the equivalence of (1) and (2) follows by applying Algebra, Lemmas 0561 and 0564. The equivalence of (2) and (3) follows by choosing a presentation $A' = R[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$ and using Algebra, Lemma 0564 to show that M is finitely presented as A'-module if and only if M is finitely presented as a $R[x_1, \ldots, x_n]$-module. □